

DATA TYPES

Data type determines the type of value we are going to store in our computer. To store anything in our computer, we should have to allocate the memory. This memory allocation is depended on the data type we are using.

Data type determines the properties such as

1. No of bytes
2. Range
3. Type of value

In C language we are having 3 **basic** data types

1. **Int – To store non-decimal numbers**
2. **Float – To store decimal numbers**
3. **Char – To stores alphabets, numbers and special char**

Total data types are divided into 3 types.

1. **Primitive data types**
2. **Derived data types**
3. **User defined data types**

PRIMITIVE DATA TYPES:

These are the regular data types we are using in our c programs.

Data type	Bytes	Conversion	Storage Range
-----------	-------	------------	---------------

		Character / format specifier	
int / signed int / short int	2	%d	-32768 to +32767
unsigned int	2	%u	0 to 65535
long int	4	%ld	-2147483648 to 2147483647
unsigned long int	4	%lu	0 to 4294967295
float	4	%f	$3.4 * 10^{-38}$ to $3.4 * 10^{+38}$
double	8	%lf	$1.7 * 10^{-308}$ to $1.7 * 10^{+308}$
long double	10	%Lf	$3.4 * 10^{-4932}$ to $1.1 * 10^{+4932}$
char	1	%c	1 character Signed char [-128 to +127] Unsigned char [0 to 255]
char[10] (STRING)	10	%s	9 char + 1 null char
void [empty data type]			nothing

DERIVED DATA TYPES:

They are derived from primitive data types.

- 1. Array [non-primitive]**
- 2. Pointer**
- 3. Function**

USER DEFINED DATA TYPES:

These are the data types created by the user.

1. structure
2. union
3. enum

String:

1. A group of characters is called string.
2. It is alpha-numeric. i.e. we can store both alphabets, numbers and special characters.

```
Char city[ ]="Hyd-16";
```

Notes:

1. One byte should be left for null char. Otherwise we are getting garbage / junk values.
2. String variable size never smaller than string. Otherwise we are getting error.
3. In C & C++ We can't copy a string with = operator. We have to use strcpy().
4. In C and C++ we can't compare two strings with == operator. We have to use strcmp().

```
char city[ ]="Hyd-16";
```

bytes



OPERATORS

Operator is a special symbol designed for a particular task. C comes with 44 operators, 14 separators. Operator works on operands. Based on no of operators participating in operation, the operators divided into 3 types.

1. **Unary operator:** Require one operand.
Eg: `a++`, `a--`, `sizeof(a)`, `~a`, `!a`, `+a`, `-a`,....
2. **Binary operators:** Require two operands.
Eg: `a+b`, `a-b`, `a<b`, `a>=b`, `a!=n`, `a<<b`,....
3. **Ternary / conditional operator:** Require 3 operands.
Eg: `a > b ? "a is big" : "b is big";`

Based on operation, the operators divided into several types.

1. **Assignment operator** `[ = ]` : It copies the value on its right side into the variable on its left side. Left side

operand should be a variable i.e. constant or expressions not allowed on left side.

Eg:

a=100;

b=1.2;

c='X';

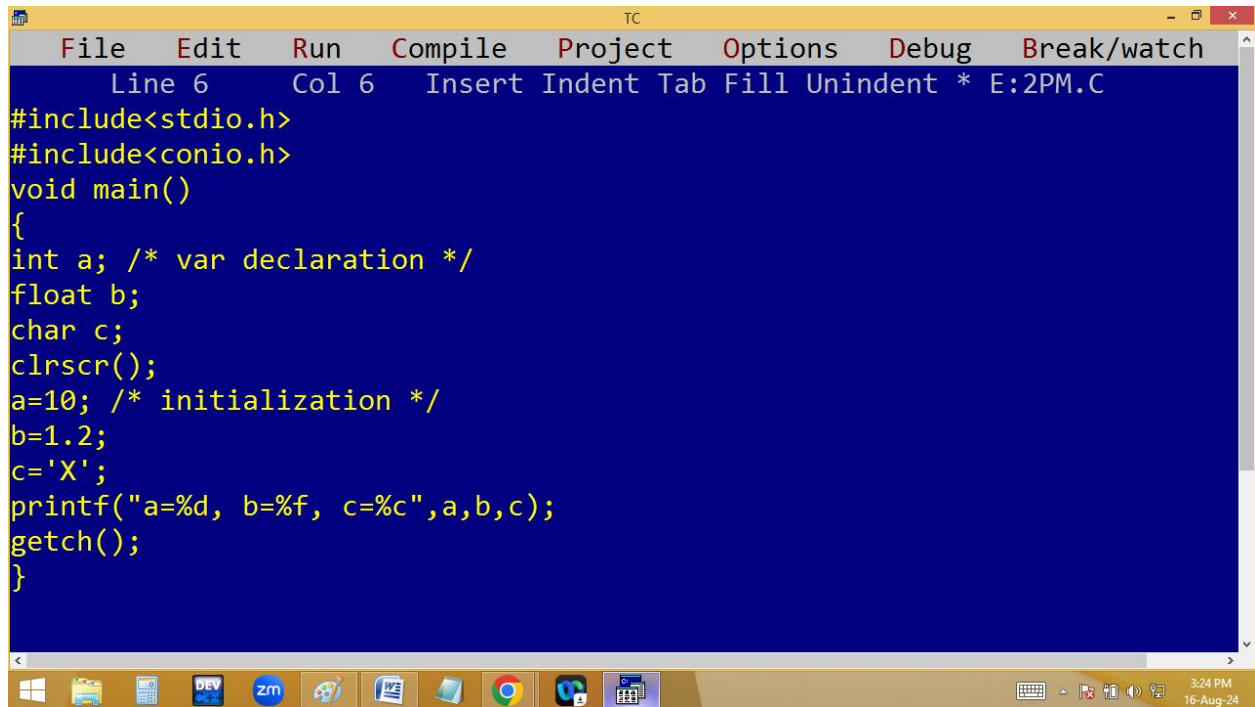
d="xyz"; → Error because of xyz is a string

e=f=g=100;

10=20; → Error because of 10 is a constant

c=10+20;

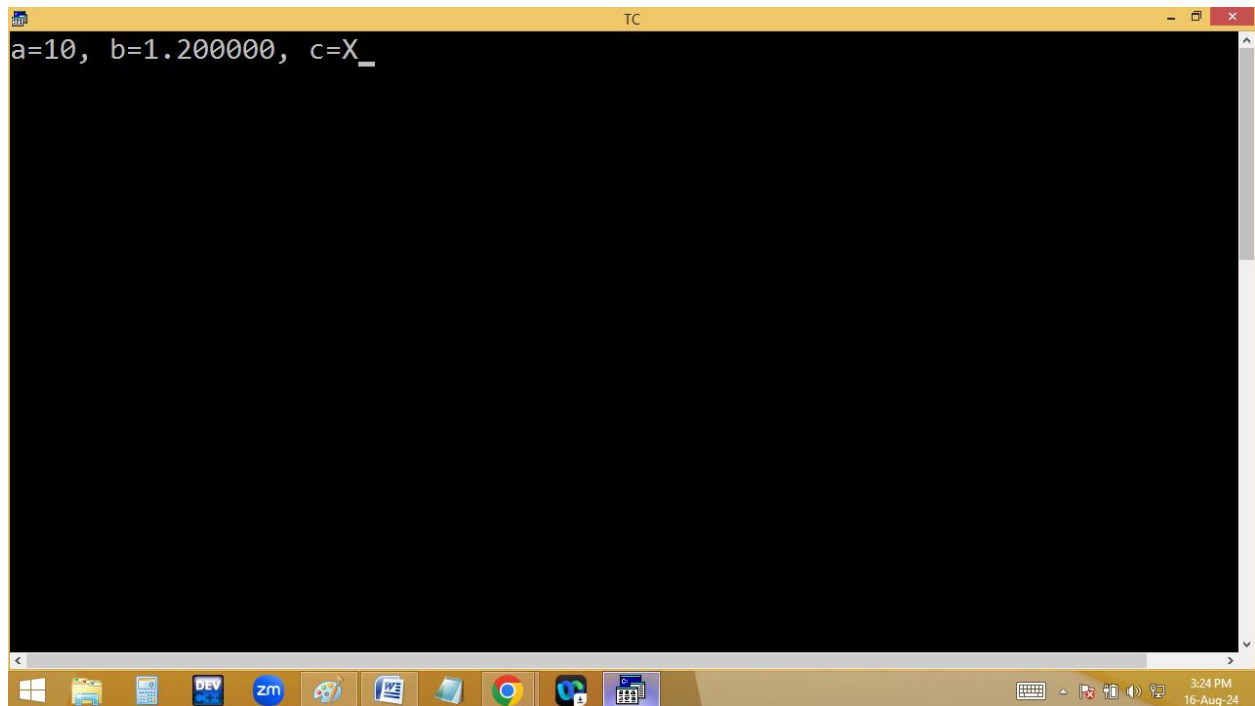
10+20=30 → **30=30;** → error



The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar and menu bar. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 6', 'Col 6', and the file path 'E:\2PM.C'. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a; /* var declaration */
float b;
char c;
clrscr();
a=10; /* initialization */
b=1.2;
c='X';
printf("a=%d, b=%f, c=%c",a,b,c);
getch();
}
```

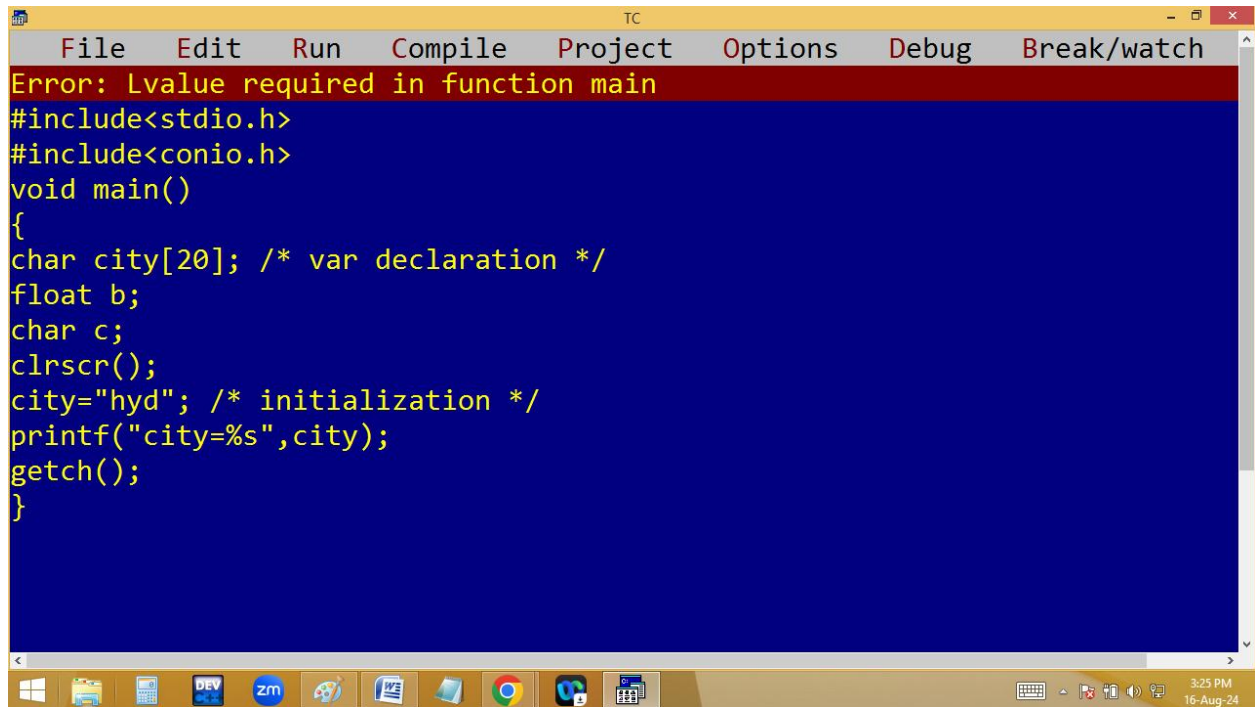
The Windows taskbar at the bottom shows various application icons and the system clock displaying 3:24 PM on 16-Aug-24.



The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar. The main editing area has a black background and displays the output of the program:

```
a=10, b=1.200000, c=X_
```

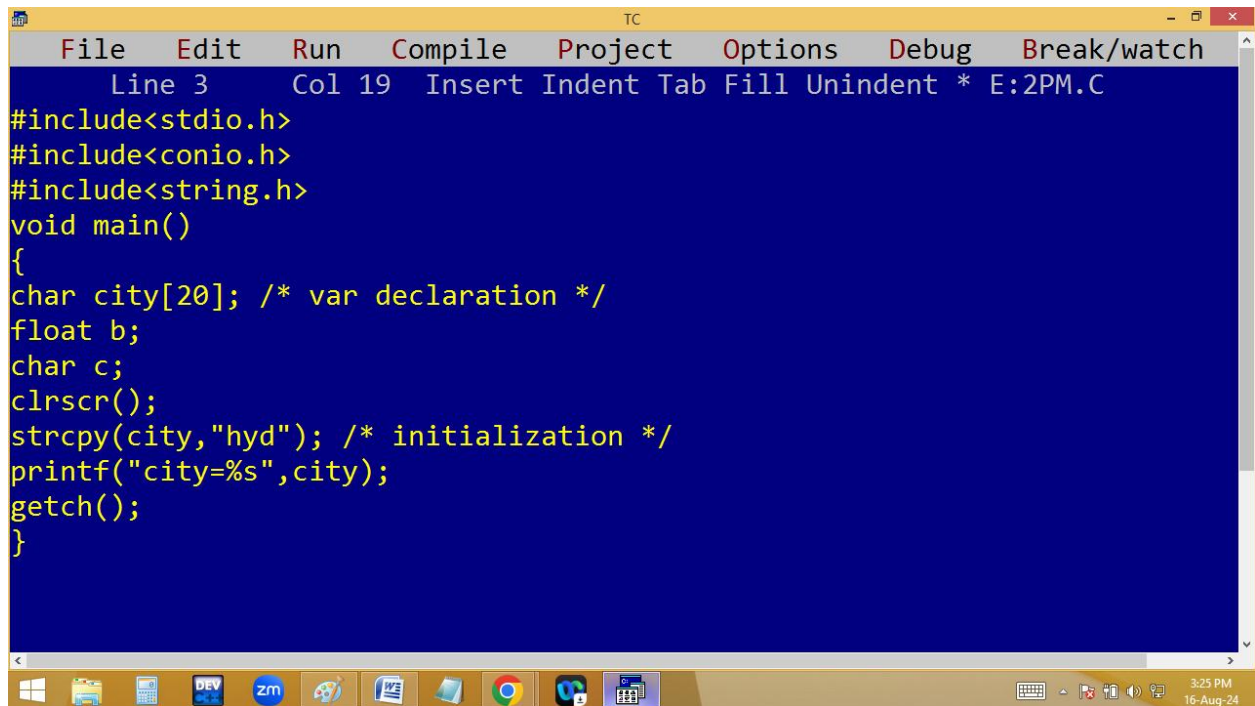
The Windows taskbar at the bottom is identical to the first screenshot, showing the same application icons and system clock.



The screenshot shows the Turbo C++ (TC) IDE with a yellow error banner at the top that reads "Error: Lvalue required in function main". The code in the editor is as follows:

```
File Edit Run Compile Project Options Debug Break/watch
Error: Lvalue required in function main
#include<stdio.h>
#include<conio.h>
void main()
{
char city[20]; /* var declaration */
float b;
char c;
clrscr();
city="hyd"; /* initialization */
printf("city=%s",city);
getch();
}
```

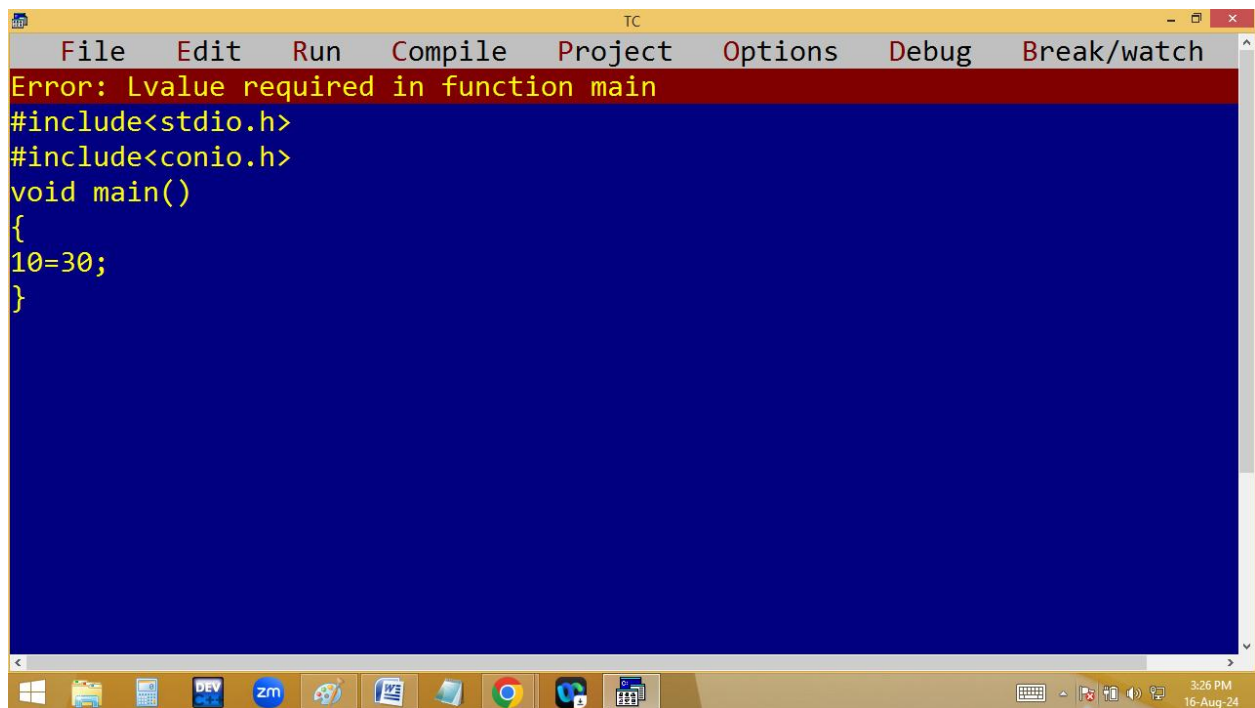
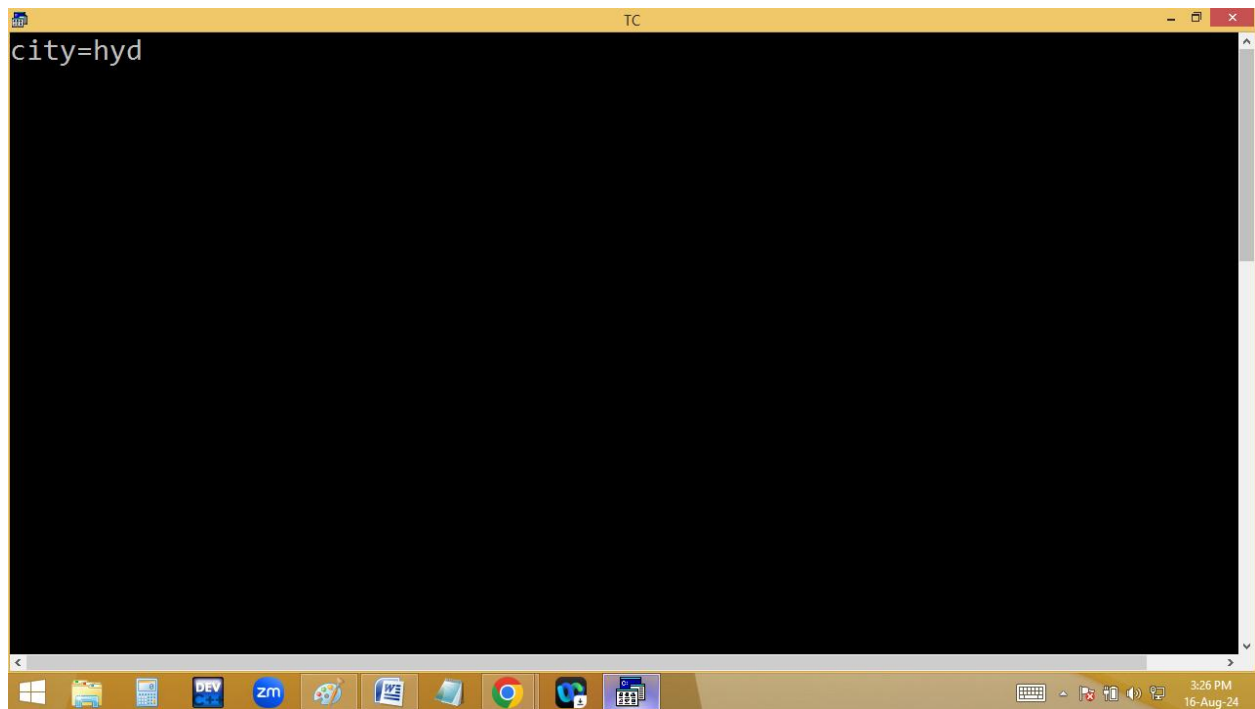
The Windows taskbar at the bottom shows the time as 3:25 PM on 16-Aug-24.

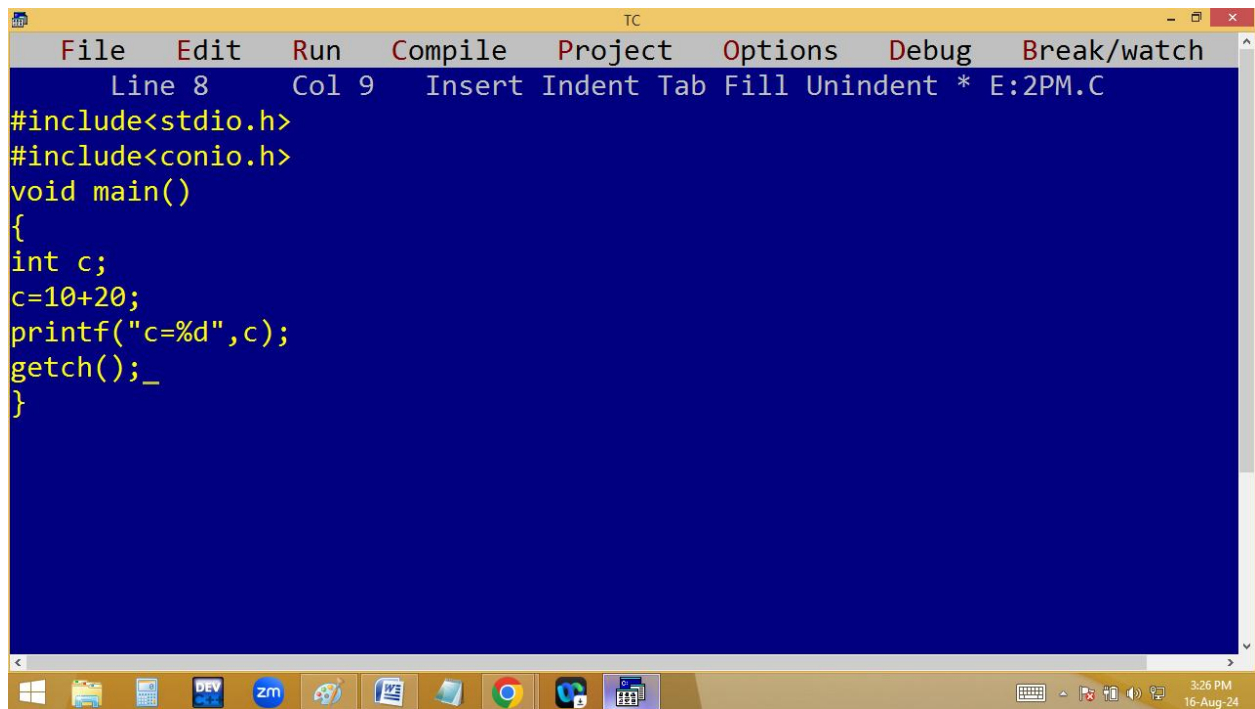


The screenshot shows the Turbo C++ (TC) IDE with the error resolved. The menu bar now includes "Line 3 Col 19 Insert Indent Tab Fill Unindent * E:2PM.C". The code in the editor is as follows:

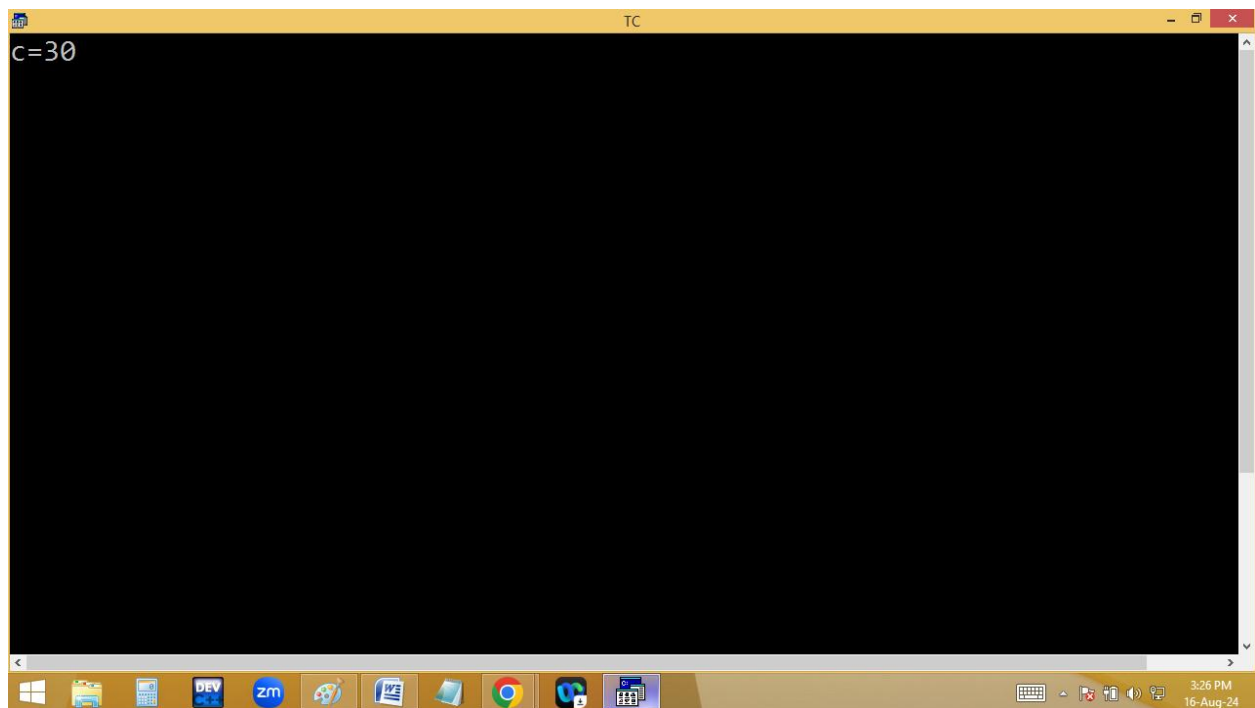
```
File Edit Run Compile Project Options Debug Break/watch
Line 3 Col 19 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
char city[20]; /* var declaration */
float b;
char c;
clrscr();
strcpy(city,"hyd"); /* initialization */
printf("city=%s",city);
getch();
}
```

The Windows taskbar at the bottom shows the time as 3:25 PM on 16-Aug-24.

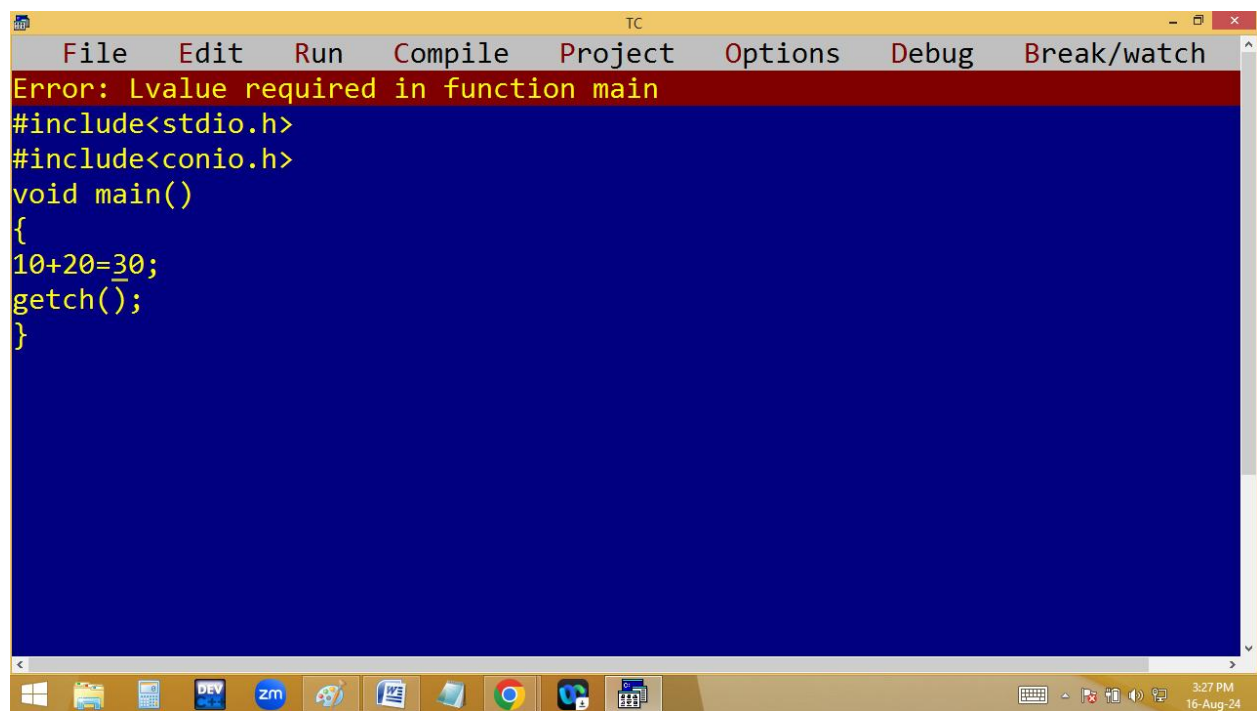




```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 9 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int c;
c=10+20;
printf("c=%d",c);
getch();_
}
```

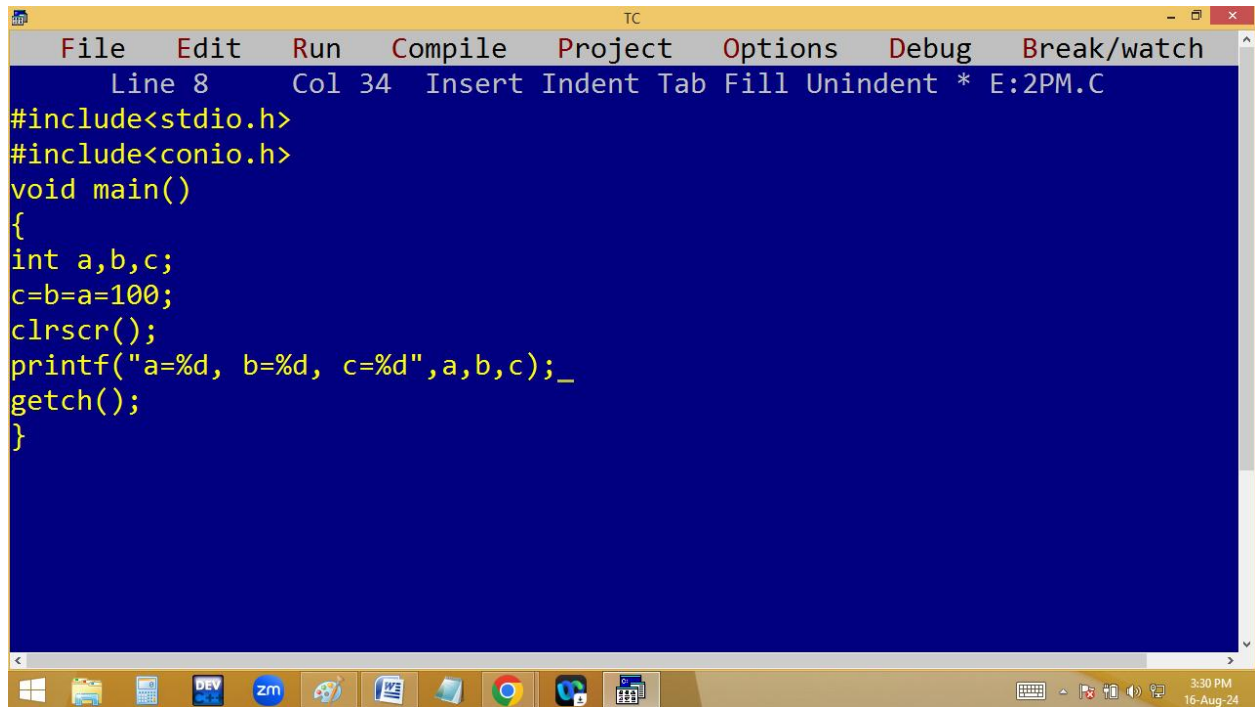


```
TC
c=30
```

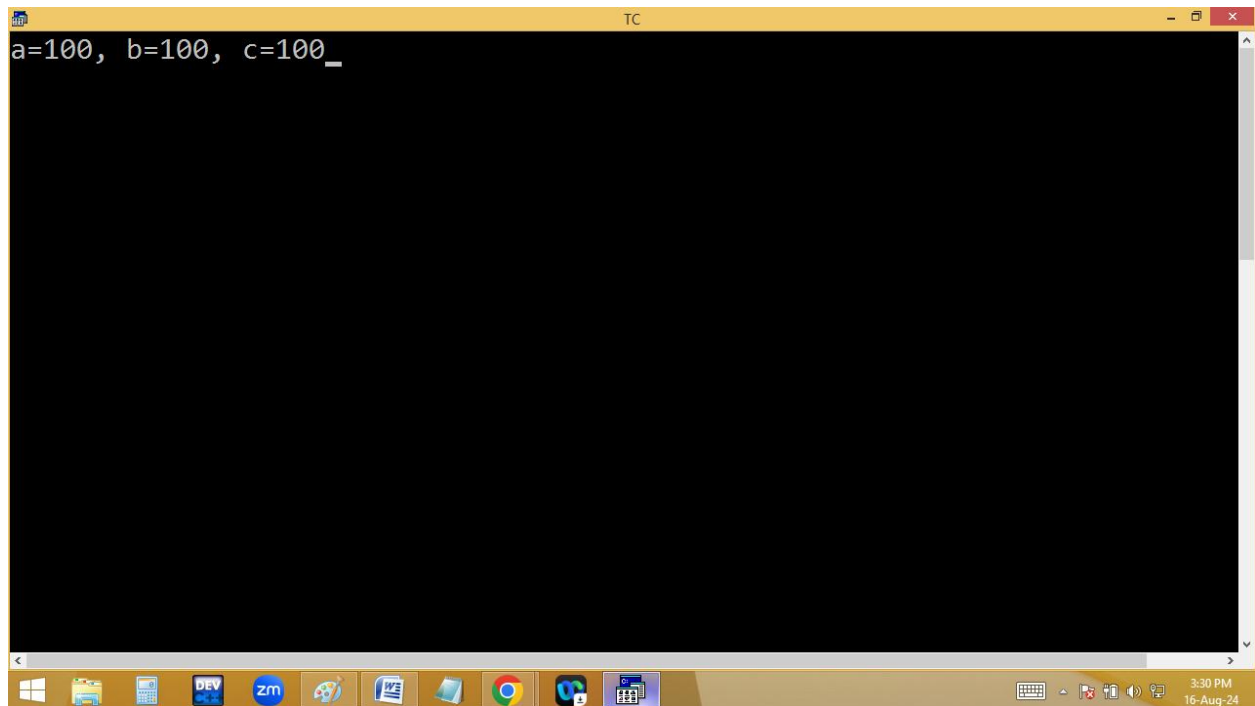


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/watch". A red error message banner at the top reads "Error: Lvalue required in function main". Below this, the source code is displayed on a dark blue background with yellow text. The code includes headers for `stdio.h` and `conio.h`, and defines a `main` function. Inside the function, there is an assignment `10+20=30;` which is causing the error, followed by a `getch();` call and a closing brace. The Windows taskbar at the bottom shows various application icons and the system clock indicating 3:27 PM on 16-Aug-24.

```
File Edit Run Compile Project Options Debug Break/watch
Error: Lvalue required in function main
#include<stdio.h>
#include<conio.h>
void main()
{
10+20=30;
getch();
}
```



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 34 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,c;
c=b=a=100;
clrscr();
printf("a=%d, b=%d, c=%d",a,b,c);_
getch();
}
```

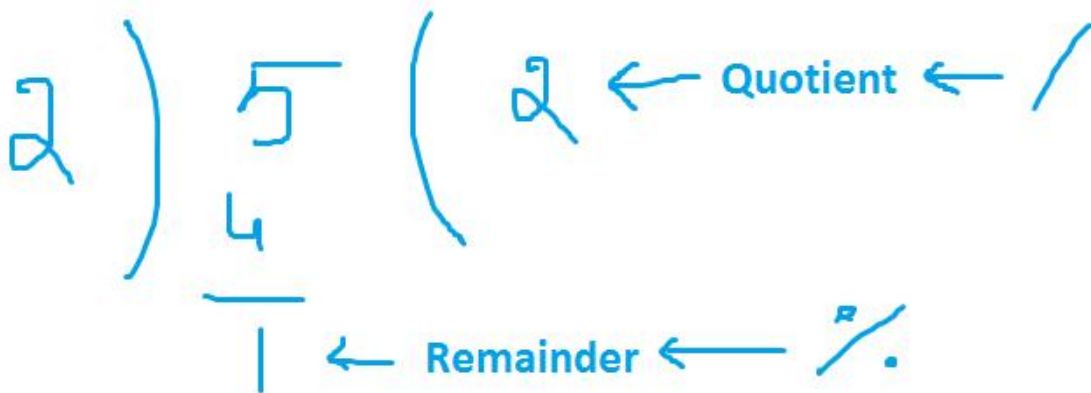


```
TC
a=100, b=100, c=100_
```

Arithmetic operators [+, -, *, %, /]: They are used to perform mathematical calculations.

Eg: $a+b$, $a-b$, $a*b$,....

% - modules [remainder]:



Handwritten long division of 5 by 2:

$$\begin{array}{r} 2 \overline{) 5} \\ \underline{4} \\ 1 \end{array}$$

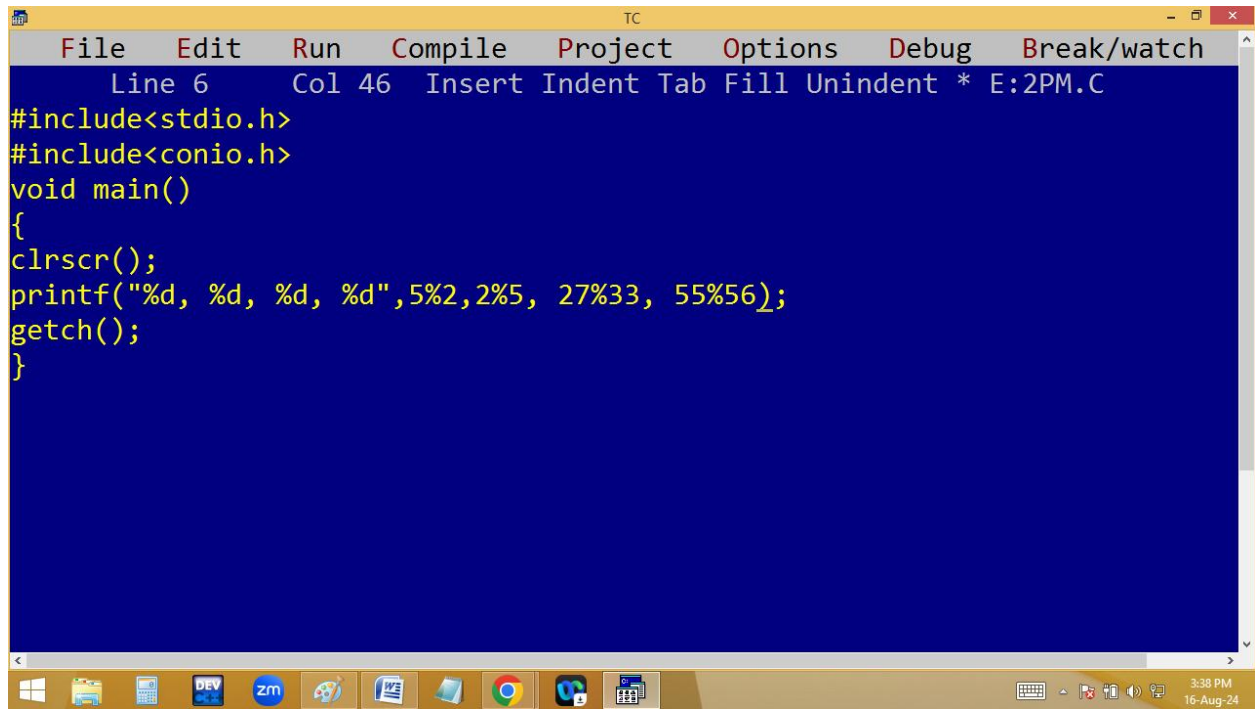
Annotations:

- 2 ← Quotient ← /
- 1 ← Remainder ← %

$$5\%2=1$$

$$2\%5=2$$

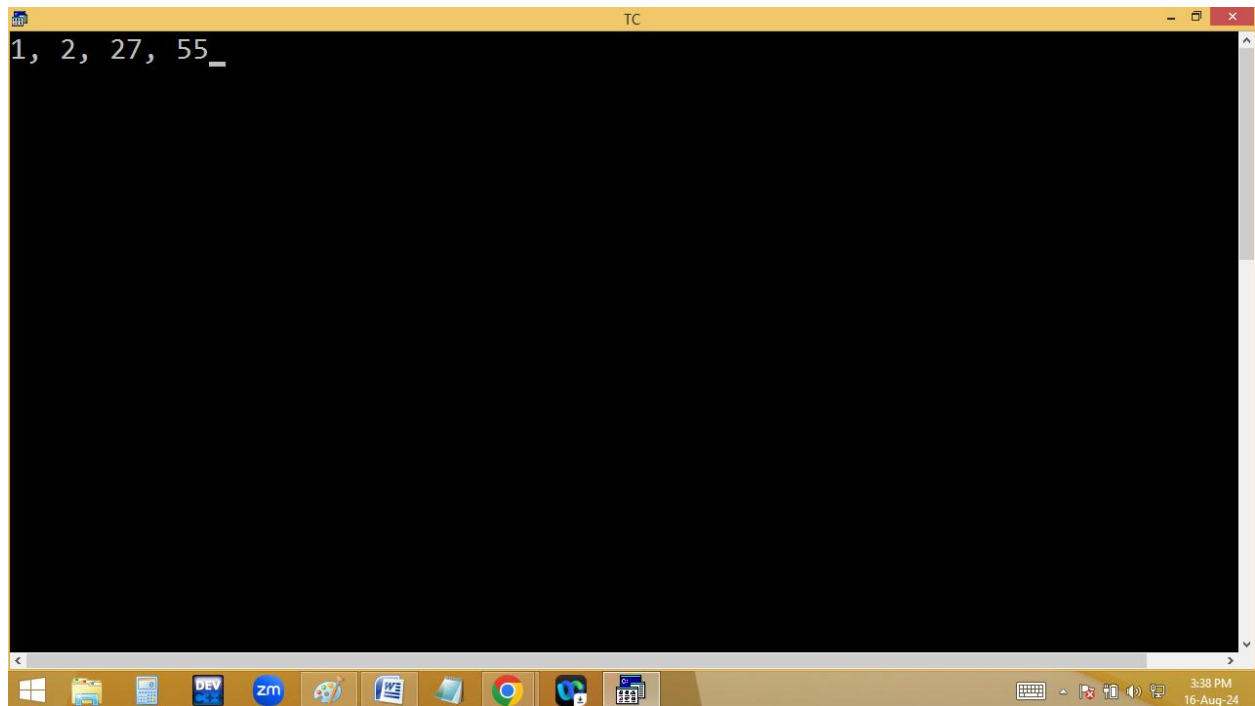
Note: If divisor bigger than dividend then dividend is the answer.



The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar and menu bar. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 6 Col 46 Insert Indent Tab Fill Unindent * E:2PM.C'. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d, %d, %d, %d",5%2,2%5, 27%33, 55%56);
getch();
}
```

The Windows taskbar at the bottom shows various application icons and the system clock indicating 3:38 PM on 16-Aug-24.



The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar. The main editing area has a black background and displays the output of the program:

```
1, 2, 27, 55_
```

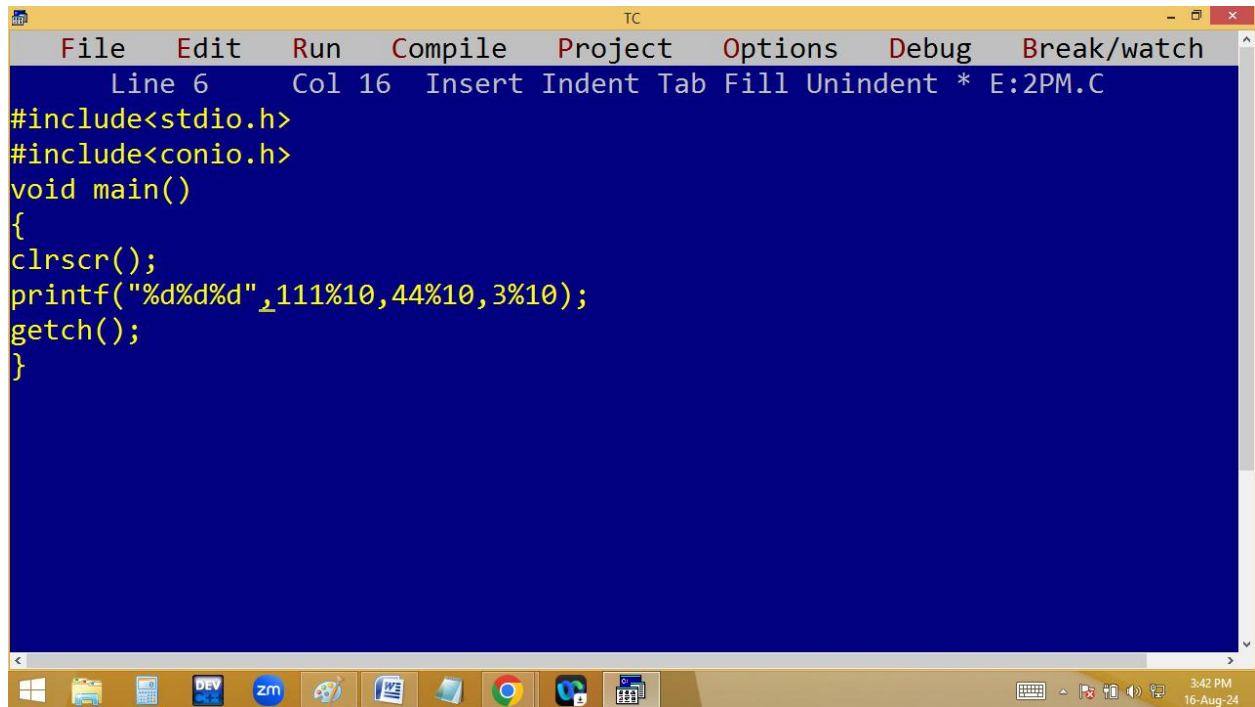
The Windows taskbar at the bottom is identical to the first screenshot, showing the same application icons and system clock.

$111 \% 10 = 1$

$444 \% 10 = 4$

$333 \% 10 = 3$

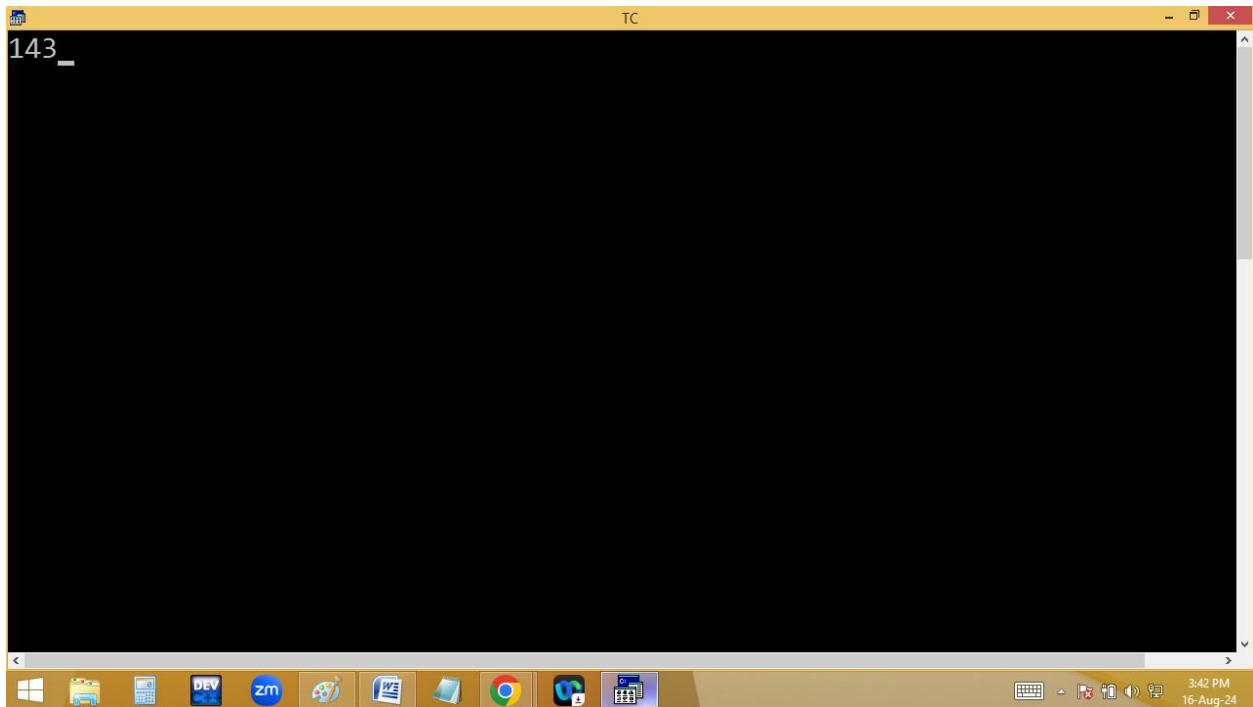
Note: Any $no \% 10$ gives last digit.



The screenshot shows a Turbo C++ (TC) IDE window. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 6 Col 16 Insert Indent Tab Fill Unindent * E:2PM.C'. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d%d%d",111%10,44%10,3%10);
getch();
}
```

The Windows taskbar at the bottom shows various icons including the Start button, File Explorer, Calculator, DEV C++, Zoom, and several other applications. The system clock in the bottom right corner displays '3:42 PM' and '16-Aug-24'.



$$\begin{array}{r} 10 \overline{) 1111} \\ \underline{110} \\ 1 \end{array}$$

$$\begin{array}{r} 10 \overline{) 141} \\ \underline{10} \\ 4 \end{array}$$

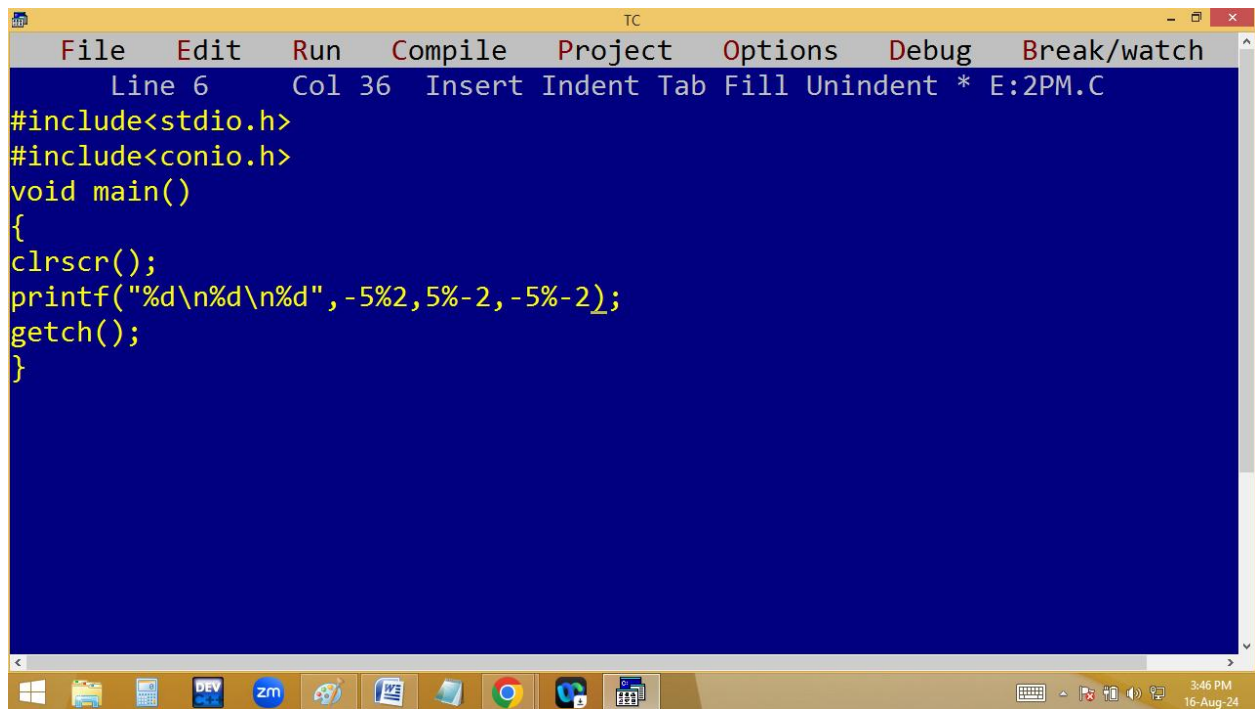
$$\begin{array}{r} 10 \overline{) 3} \\ \downarrow \\ 3 \end{array}$$

$$-5 \% 2 = -1$$

$$5 \% -2 = 1$$

$$-5 \% -2 = -1$$

Note: In modules if the numerator is negative then result also negative.



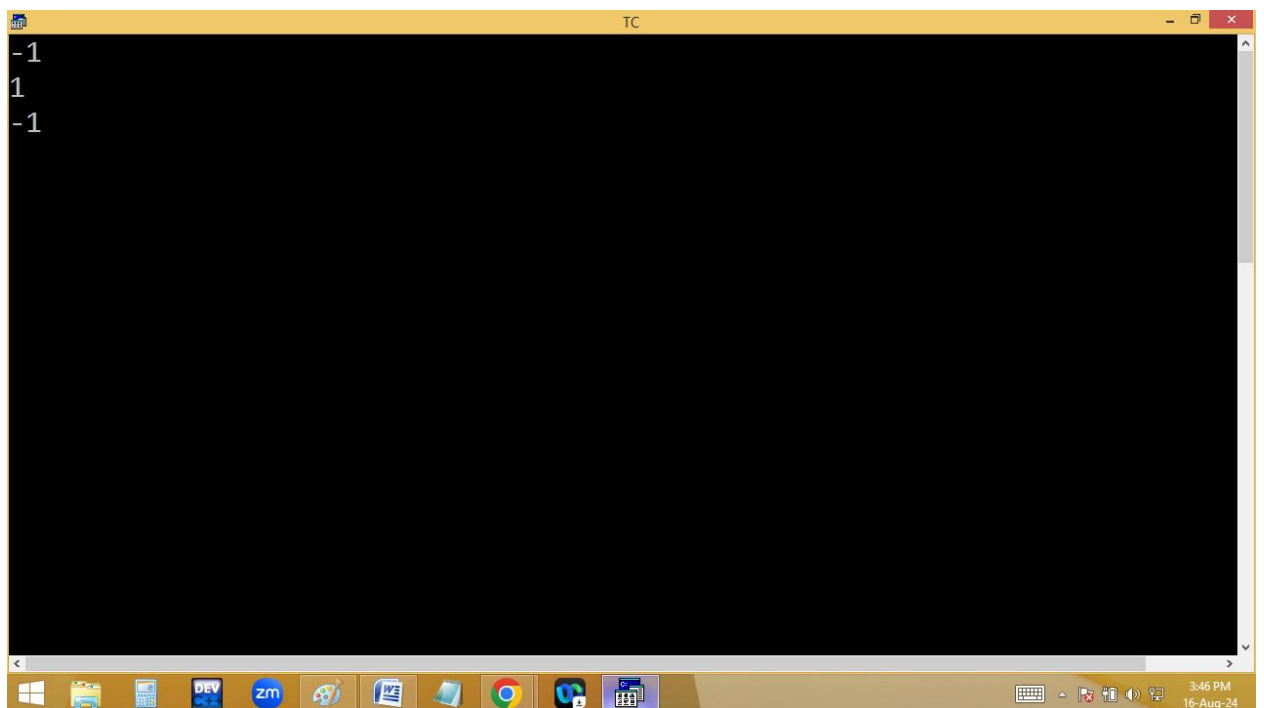
TC

File Edit Run Compile Project Options Debug Break/watch

Line 6 Col 36 Insert Indent Tab Fill Unindent * E:2PM.C

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n%d\n%d", -5%2, 5%-2, -5%-2);
getch();
}
```

Windows taskbar with icons for DEV, zm, and other applications. System clock shows 3:46 PM on 16-Aug-24.



TC

```
-1
1
-1
```

Windows taskbar with icons for DEV, zm, and other applications. System clock shows 3:46 PM on 16-Aug-24.

